

LINUX LAB-1

Que-1

Que 1: Create a LabFiles directory and perform create/copy/move/rename/delete operations using touch, cp, mv, rm, unlink, ls, stat.

Ans 1:

Theory

Linux provides several commands for managing files and directories. `mkdir` creates a directory, while `touch` creates empty files. `ls` displays directory contents. `cp` copies files, whereas `mv` is used to move or rename files. `stat` displays detailed file metadata such as size, permissions and timestamps. `rm` deletes files, while `unlink` removes a single file or hard link. These commands are used to perform basic file management operations in Linux.

Result

The LabFiles directory was created successfully, and file creation, copying, renaming, metadata checking, deletion and hard link removal operations were performed successfully using Linux commands.

```
root@mshuaibsi:/home/LabF x + v
root@mshuaibsi:/home# mkdir LabFiles && cd LabFiles
root@mshuaibsi:/home/LabFiles# touch file1.txt file2.txt
root@mshuaibsi:/home/LabFiles# ls -l
total 0
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file2.txt
root@mshuaibsi:/home/LabFiles# cp file1.txt file1_copy.txt
root@mshuaibsi:/home/LabFiles# ls -l
total 0
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1_copy.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file2.txt
root@mshuaibsi:/home/LabFiles# mv file2.txt file2_renamed.txt
root@mshuaibsi:/home/LabFiles# ls -l
total 0
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1_copy.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file2_renamed.txt
root@mshuaibsi:/home/LabFiles# stat file1_copy.txt
  File: file1_copy.txt
  size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,48      Inode: 2440         Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2026-09-19 15:53:32.570587518 +0000
Modify: 2026-09-19 15:53:32.570587518 +0000
Change: 2026-09-19 15:53:32.570587518 +0000
 Birth: 2026-09-19 15:53:32.570587518 +0000
root@mshuaibsi:/home/LabFiles# rm file1_copy.txt
root@mshuaibsi:/home/LabFiles# ls -l
total 0
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file2_renamed.txt
root@mshuaibsi:/home/LabFiles# ln file2_renamed.txt file2_hardlink.txt
root@mshuaibsi:/home/LabFiles# unlink file2_hardlink.txt
root@mshuaibsi:/home/LabFiles# ls -l
total 0
-rw-r--r-- 1 root root 0 Sep 19 15:53 file1.txt
-rw-r--r-- 1 root root 0 Sep 19 15:53 file2_renamed.txt
root@mshuaibsi:/home/LabFiles# |
```